STUDY NOTES · THE JOINT MODEL

Joint alignment, question by question

The questions you asked while working through the paper, in the order you asked them. Each answer goes intuition first, then a picture, then the maths.

Every answer was checked against the paper and the code. Where the chat got something wrong, the slide says so, and two slides at the end collect the corrections.

YOUR QUESTIONS

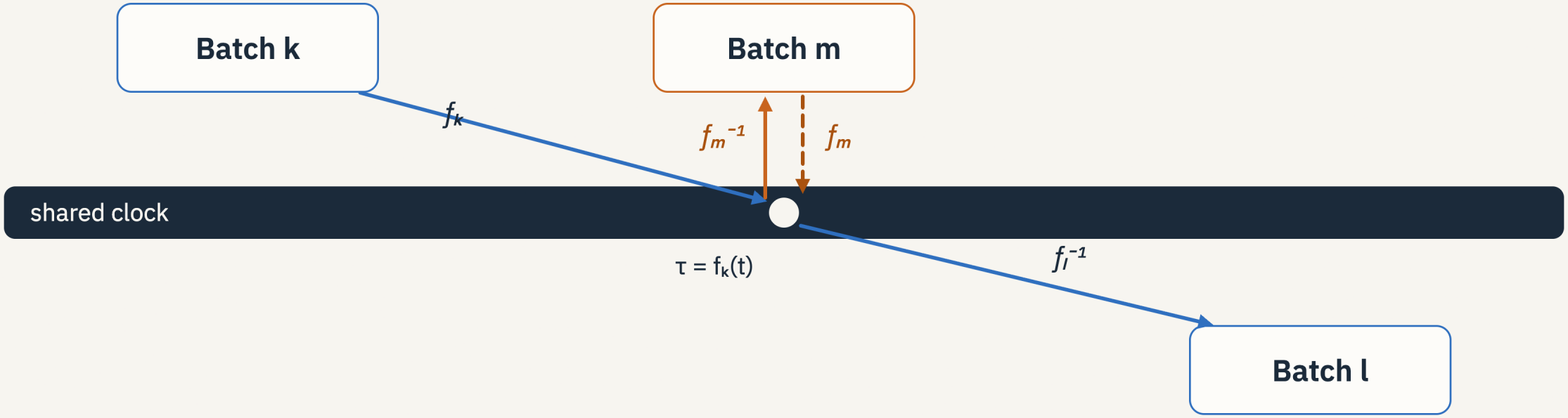
Fourteen questions, in the order you asked them

1 Why does every route between batches give the same map?
2 Why does fitting never need an inverse?
3 What is a piecewise-linear hat basis?
4 Basis weights or knot parameters: which is the warp?
5 Why store increments instead of knot values?
6 What does min ‖Aθ − y‖² subject to bounds say?
7 Is the loss “strictly convex”?
8 What does “positive definite” mean?
9 What is the gauge, and does the choice matter?
10 How does robust reweighting (IRLS) work?
11 Is IRLS like Adam?
12 Why can’t ordinary cross-validation pick the smoothness?
13 What do the band and the interval tell us, and why 81%?
14 How does one batch borrow strength from the others?

Each slide’s eyebrow gives the question number and whether it is the intuition or the maths.

Q1 · INTUITION

A detour steps off the clock and back on



Pairwise fitting gives every pair its own map from its own anchors: 10 maps for 5 batches. Nothing forces k → m → l to agree with k → l.

The joint model gives each batch one warp onto a shared clock. Going via m means stepping off at τ and back on at the same τ, so the detour changes nothing.

You asked: how does “transitivity by construction” hold for any two paths between batches?

Q1 · MATHS

Every path telescopes to the same map

Define the map from batch k to batch l as $g_{k\to l} = f_l^{-1} \circ f_k$: on with f_k, off with f_l^{-1}. Then for any m:

$$g_{m\to l} \circ g_{k\to m} = f_l^{-1} \circ (f_m \circ f_m^{-1}) \circ f_k = f_l^{-1} \circ f_k = g_{k\to l}$$

Any longer path $k \to p_1 \to p_2 \to l$: every inner pair cancels to the identity.

$$f_l^{-1} \quad f_{p2} \circ f_{p2}^{-1} = \mathrm{id} \quad f_{p1} \circ f_{p1}^{-1} = \mathrm{id} \quad f_k = g_{k\to l}$$

On the fish data

Composed pairwise maps miss the direct map by 0.116 min on average over 60 ordered triplets, and by up to 0.277 min, more than one bin. Joint model: at most 0.0023 min, from inverting on a grid.

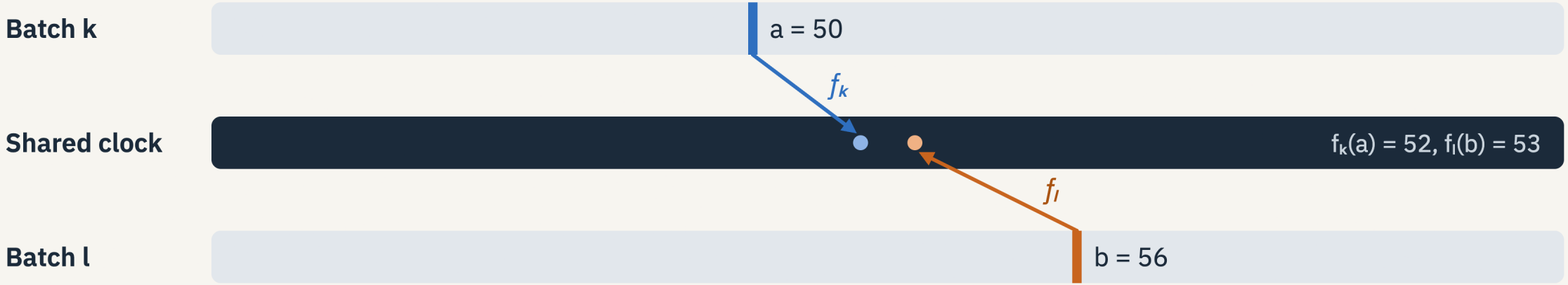
Consistent is not correct

The cancelling only needs each f_k to have an inverse; the slope bounds guarantee that. If an f_k is wrong, every map through it is wrong in the same way.

Paper Eq. (1) and the transitivity figure. Function composition is associative, which is what lets the brackets move.

Q2 · INTUITION

Fitting only ever warps forward



`miss r = fₖ(a) - fₗ(b) = 52 - 53 = -1 bin`

While fitting

Both peaks go **forward** onto the clock and we score the gap. No inverse, so each miss stays linear in the unknowns.

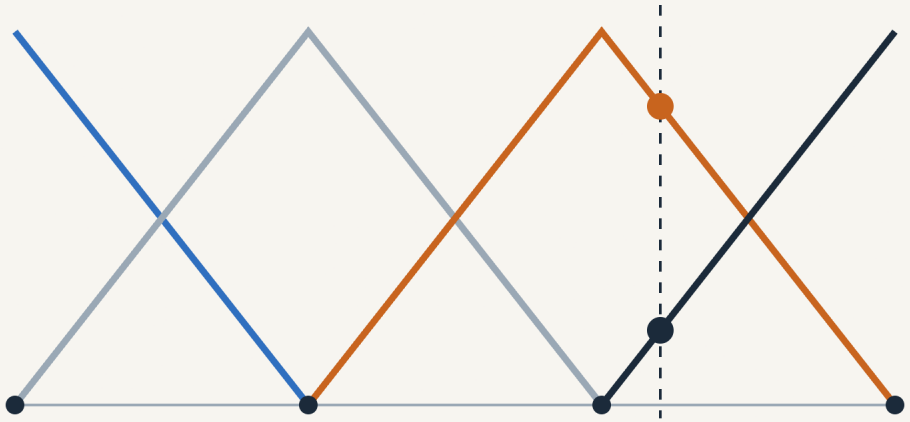
After fitting

To move a whole chromatogram from k to l we need $g_{k\to l} = f_l^{-1} \circ f_k$. The inverse is computed once, by interpolation.

You said: “We only care about the residual … not their warps back (the inverse) to another batch’s clock.” Correct. Toy numbers: $d_k(50) = +2$, $d_l(56) = -3$.

Q3 · INTUITION AND MATHS

Each knot wears a hat; the two nearest share t



Knots 0–3 at t = 0, 10, 20, 30. Each hat is 1 on its own knot and 0 at its neighbours. Dashed: t = 22 meets knot 2's hat at 0.8 and knot 3's at 0.2.

For t between a left and a right knot, Δu apart:

```
left knot: (right - t) / Δu
right knot: (t - left) / Δu
```

They add to 1. Stack all the hats' heights at t into a row: **B(t)**. At most two entries are non-zero, and exactly one when t sits on a knot.

`B(22) = (0, 0, 0.8, 0.2)`

Your summary ("at most two knots, or one knot if we are directly over a knot") is exactly right. The model uses Q = 41 knots, about 1.1 min apart.

Q4 · MATHS

The warp needs both: weights times values

FIXED · GEOMETRY

Basis weights B(t)

How close is t to each knot? Known from t and the knot grid before fitting. Never learned.

LEARNED · DATA

Knot values $L\theta_k$

How much has batch k drifted at each knot? These are what the solver fits, one set per batch.

$$B(22) = (0,\ 0,\ 0.8,\ 0.2) \cdot L\theta_k = (0,\ 2,\ 1,\ 3) = 0.8 \cdot 1 + 0.2 \cdot 3 = 1.4 = d_k(22)$$

The dot product is the drift at t, and the warp adds t back: **$f_k(t) = t + B(t)\ L\ \theta_k$**. The knot values say *what* the drift is; the weights say *how* to read it between knots.

You asked: is the warp stored in the weights, the parameters, or the product of both? The product. And yes, B(t) is the row of basis weights.

Q4 · MATHS, CONTINUED

One anchor becomes one row of a big matrix

$$f_k(a) = a + B(a)\ L\ \theta_k$$
$$f_l(b) = b + B(b)\ L\ \theta_l$$
$$r = f_k(a) - f_l(b) = (a - b) + B(a)\ L\ \theta_k - B(b)\ L\ \theta_l$$

The row for this anchor, across all five batches' parameters (41 columns each):

0 … 0	+B(a)L (batch k)	0 … 0	-B(b)L (batch l)	0 … 0

With target **y = b − a**, each miss is **r = (row · θ) − y**. The orange term is a known number and the blue term is linear in θ, so stacking every anchor gives **r = Aθ − y**: on the fish data, 25,011 rows by 205 columns.

You wrote "a − b + B(a)Lθ$_k$ − B(b)Lθ…": correct, with θ$_l$ at the end. Paper Eq. (3); code builds A as X with y = tb − ta.

Q5 · INTUITION AND MATHS

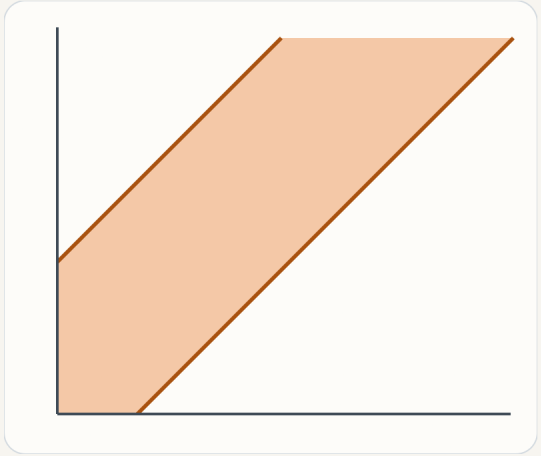
Increments turn the slope rule into a box

$\theta_k = (c, \delta_1, \delta_2, \ldots)$: the drift at the first knot, then the step to each next knot. L adds the steps up: $\mathbf{L\theta_k}$ = knot values.

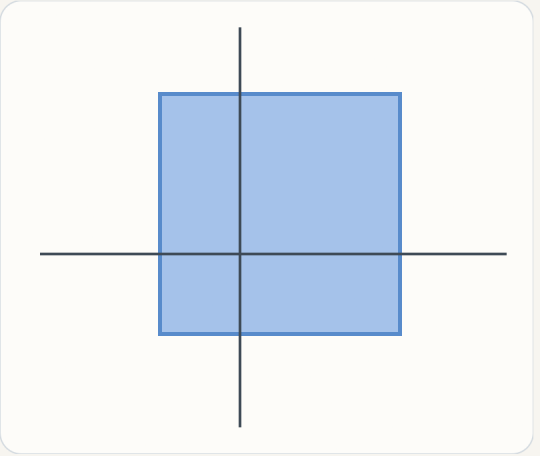
The warp's slope between knots is $1 + \delta/\Delta u$. Keeping it in $[0.5, 2]$:

$$-0.5 \cdot \Delta u \le \delta \le 1.0 \cdot \Delta u$$

With $\Delta u \approx 4.98$ bins: each step lies in $[-2.49, 4.98]$ bins. The first entry, c, has no bound.



Knot values v_q, v_q+1: the rule ties neighbours together, a slanted band.



Increments δ_q, δ_q+1: each has its own interval, a box.

A box needs only **lower and upper bounds per unknown**, which bounded-variable least squares handles directly. A band would need a general constrained solver.

You asked: is L just a convenient way to sum the increments? Yes, and that choice is what makes the bounds simple. Paper Eq. (2); $\Delta u = 199/40$ bins.

Q6 · MATHS

Reading the bounded least-squares problem

$$\min_{\theta} \|A\theta - y\|^2 \text{ subject to } lb \le \theta \le ub$$

θ: every unknown

All batches' $(c, \delta_1, \ldots, \delta_{40})$ stacked: $5 \times 41 = 205$ numbers.

A and y: the anchors

One row per anchor (previous slide); y holds each anchor's raw offset b − a.

‖·‖²: add up squared misses

Aθ − y is the vector of misses r; squaring and summing scores the whole fit.

lb ≤ θ ≤ ub: the box

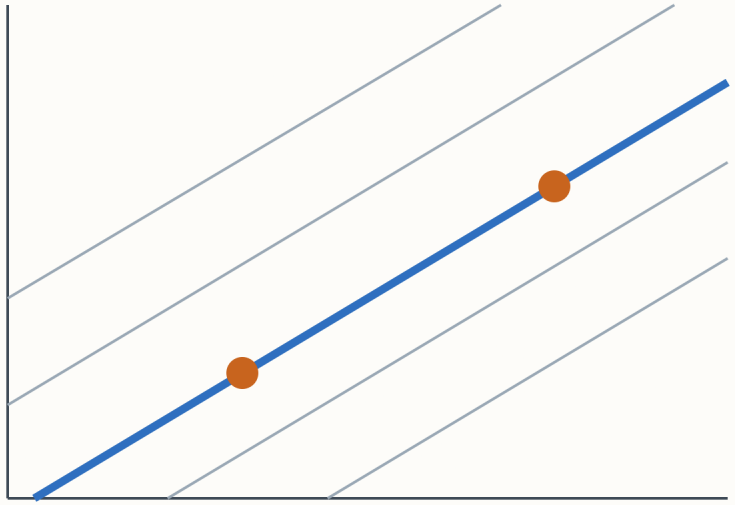
Each δ in $[-0.5\Delta u, \Delta u]$; each c free $(-\infty, \infty)$. One interval per unknown.

The full objective also has robust weights, a smoothness term and a gauge term. Each is a sum of squares of linear expressions, so each becomes **extra rows of A** (a weight w scales its row by √w). The problem keeps this exact shape.

Paper Eq. (4). In code: Cholesky of the normal matrix, then scipy's lsq_linear(method='bvls'). Stacking rows and the Cholesky route give the same answer.

Q7 · INTUITION AND MATHS

Convex, but with a flat valley floor



Two batches, one drift each (d_1 across, d_2 up). Grey: equal-loss lines. Blue: the valley floor $d_2 = d_1 - y$. Both orange points have the lowest loss.

Anchors only see the **gap** $d_1 - d_2$, so the loss is $(d_1 - d_2 - y)^2$. Slide along the valley, adding the same amount to both drifts, and the loss never changes.

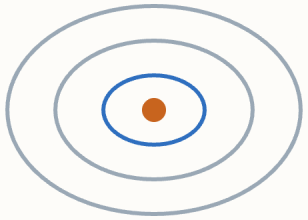
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H = 2·[[1, -1], [-1, 1]]
eigenvalues 4 and 0
0 belongs to direction (1, 1)
```

Convex: it never curves down, so every minimum is a global one. **Not strictly convex**: one direction is flat, so the minimum is not unique.

You caught this: the chat first said “strictly convex”, then corrected itself. In the real model the flat direction is “add one constant to every batch’s drift”.

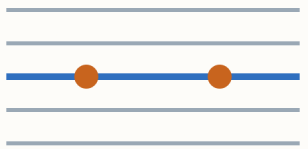
Q8 · REFRESHER

Positive definite means a bowl, not a trough



Positive definite: a bowl

$x^T Hx > 0$ for every $x \neq 0$. All eigenvalues > 0. One lowest point.



Semi-definite: a trough

$x^T Hx \geq 0$, and $= 0$ along some x. An eigenvalue is 0. A whole line of lowest points.

Back to the toy. Add a gauge term $\gamma \cdot d_1^2$ (“pin batch 1’s drift near 0”):

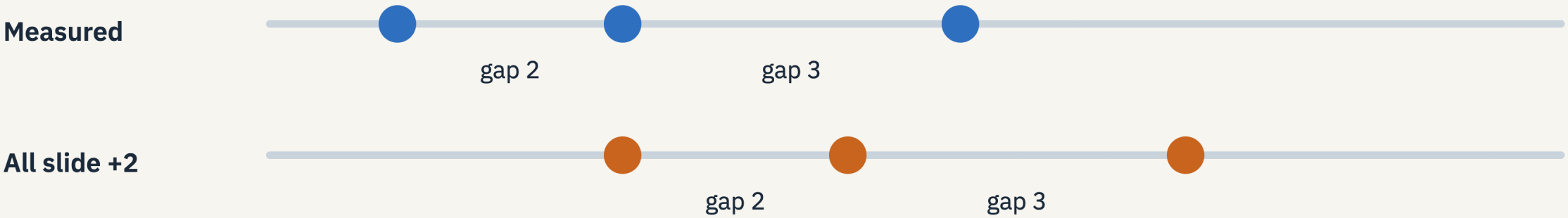
$$\tfrac{1}{2}\, x^T Hx = (x_1 - x_2)^2 + \gamma \cdot x_1^2$$

$= 0$ only if $x_1 = x_2$ and $x_1 = 0$, so only at $x = 0$ → a bowl

x is any direction you could step in; $x^T Hx$ is how sharply the loss curves that way. The code also adds a tiny $10^{-8} \cdot N_eff$ to every diagonal entry.

Q9 · INTUITION

The gauge nails the clock to the ground



Anchors measure only the **gaps** between batches, like timing runners only against each other. Move everyone 2 forward and every gap is the same, so the data cannot say where zero is. The gauge adds that one missing rule.

Reference gauge (the paper's default)

One batch is zero: L θ_r = 0. Its clock is the shared clock. Here r = sep2015, the batch with the most anchors.

Mean gauge

The drifts add to zero at every knot: Σ_k L θ_k = 0. The shared clock is the batches' average.

Added as a strong penalty γ‖Gθ‖² with γ = 10³·N_eff/Q, so the rule holds almost exactly. Paper Eq. (4).

Q9 · MATHS · A CORRECTION

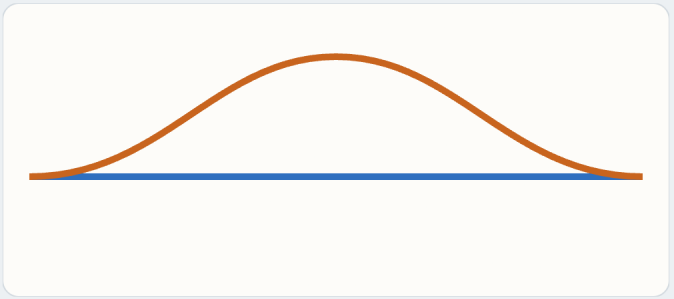
Gauges differ in the clock’s shape, not its shift

A constant shift changes nothing

Add s to every drift: misses unchanged, second differences of a constant are 0, and c has no bound. Penalty and bounds are untouched too.

A common bend does matter

Each gauge pins a whole curve (41 knot values). Bending the clock, $f_k \longrightarrow h \circ f_k$ for every k, leaves every map the same but changes each warp’s wiggle.



Reference: $d_1 = 0$ (blue), $d_2 = e$ (orange)



Mean: $d_1 = -e/2$, $d_2 = +e/2$

Same gap e between the batches, so nearly the same map. Smoothness penalty:

$$\text{ref: } \|D_2 e\|^2$$

$$\text{mean: } 2 \cdot \|D_2 e/2\|^2 = \tfrac{1}{2}\|D_2 e\|^2$$

So one λ smooths the same relative drift **half as hard** under the mean gauge in this two-batch toy. That is how the gauge interacts with the smoothness penalty.

The chat said changing the gauge “shifts” the clock and so changes the penalty. A shift cannot; a bend can. The toy is a first-order sketch, not a result from the paper.

Q9 · WORKED EXAMPLE: SHIFT VS SHAPE

Slide the ruler or stretch it: same maps, different bends

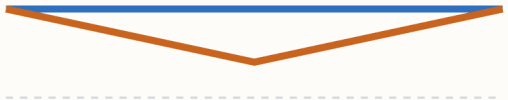
Two batches. Compounds at 10, 20, 30 in A sit at 10, **23**, 30 in B: a bump of 3 in the middle. Plot each batch's drift at those three compounds (clock time minus its own time) under three choices of clock.

Reference: clock = A



A: 0, 0, 0 · B: 0, −3, 0
Bends: 0 and 3 · **penalty $0^2 + 3^2 = 9$**

Shift: clock = A + 5



A: 5, 5, 5 · B: 5, 2, 5
Bends: 0 and 3 · **penalty 9** (unchanged)

Shape: clock = average



A: 0, +1.5, 0 · B: 0, −1.5, 0
Bends: 1.5 and 1.5 · **penalty 4.5**

In all three, A's 20 and B's 23 land on the same clock point, so **the map A → B is identical**. Sliding the clock moves every curve together and bends nothing. Re-shaping it moves bend from one warp to the other, and the penalty sees that.

Blue: batch A. Orange: batch B. Grey dashed: zero drift. "Bend" here is the bump height at the middle knot; the real penalty squares second differences, which behaves the same way.

Q9 · EVIDENCE

No gauge is correct, and the paper leaves it open

At λ = 10	Reference	Mean
Held-out anchor error (lower is better)	0.111 min	0.096 min
TIC correlation, drifted pairs	0.598	0.595
TIC correlation, near-aligned pairs	0.779	0.768

A trade-off: the mean gauge predicts held-out anchors better; the reference gauge gives slightly higher TIC correlation.

You said: no mathematically correct gauge

Right. The data cannot pick one, so the choice is a convention.

But it was not chosen by experiment

The selection rule fixes the reference gauge and tunes only λ. This table compares; it did not select. The paper calls the choice “not settled”.

Your idea that a different penalty might favour the mean gauge fits the previous slide, but it has not been tested. Paper: sensitivity results.

Q10 · INTUITION AND MATHS

Eight solves: plain, then Huber, then Tukey

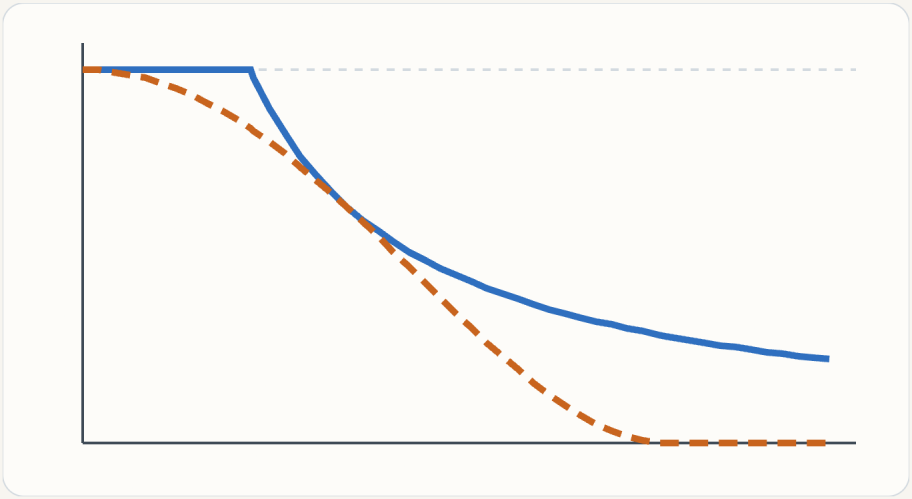
1 plain solve
every weight 1

→

3 Huber-weighted solves
shrink big misses

→

4 Tukey-weighted solves
drop gross mismatches



Weight (up, 0 to 1) against miss size |r|/σ̂ (across, 0 to 6). Blue solid: Huber. Orange dashed: Tukey, exactly 0 beyond 4.685.

$$\hat{\sigma} = \max(1.4826 \cdot \mathrm{MAD}(r),\ 0.25 \text{ bins})$$

$$\text{Huber: } w = \min(1,\ c/|r|),\quad c = 1.345\hat{\sigma}$$

$$\text{Tukey: } w = (1 - (r/c)^2)^2,\quad c = 4.685\hat{\sigma}$$

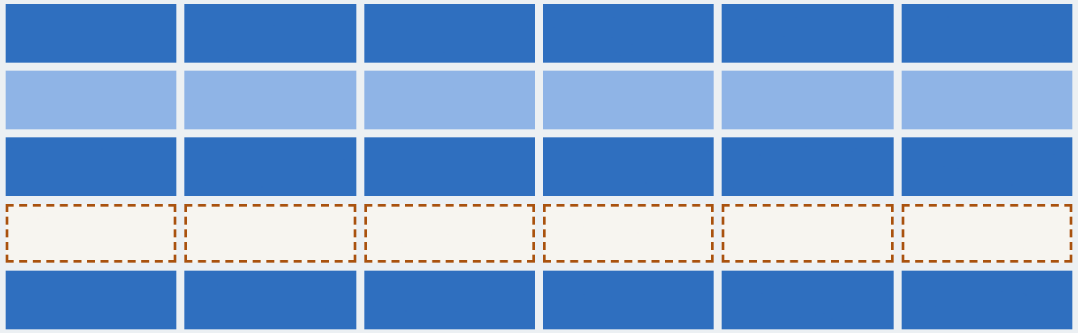
$$w = 0 \text{ when } |r| \geq c$$

Huber's loss is **convex**, so it starts safely. Tukey's is not: begun from a poor fit it could throw away good anchors. Hence Huber first.

Fixed schedule, not "until θ stops changing". Each solve has one answer; the robust fit as a whole is not convex, so its answer depends on the start. 3.9% of fish anchors end at weight 0.

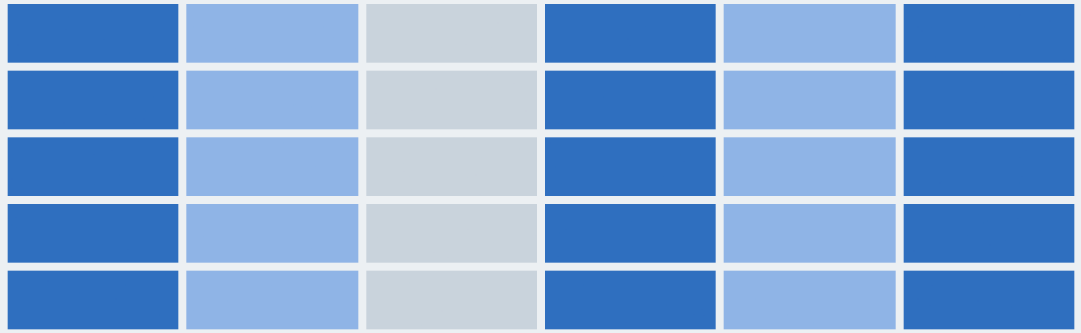
Q11 · INTUITION

IRLS weighs rows; Adam scales columns



IRLS: one weight per anchor (row)

“How much do I trust this data point?” Pale row: down-weighted. Dashed row: weight 0, a rejected mismatch. All unknowns are then solved together, exactly.



Adam: one step size per parameter (column)

“How far should I move this unknown?” Every data point counts fully; each parameter takes its own gradient step.

A closer analogy from machine learning: per-sample loss weighting, as in robust regression or down-weighting noisy labels.

Q12 · INTUITION AND MATHS

Hold out stretches of time, not samples



The 0–45 min run as 17 blocks of 12 bins (2.7 min); block b goes to fold b mod 5. Orange: fold 2, held out across every sample and batch.

Folds over samples leak

Hide one sample and its compounds still have anchors at the same times in other samples. The test never visits the gaps between compounds, where a wiggly warp goes wrong.

Folds over time do not

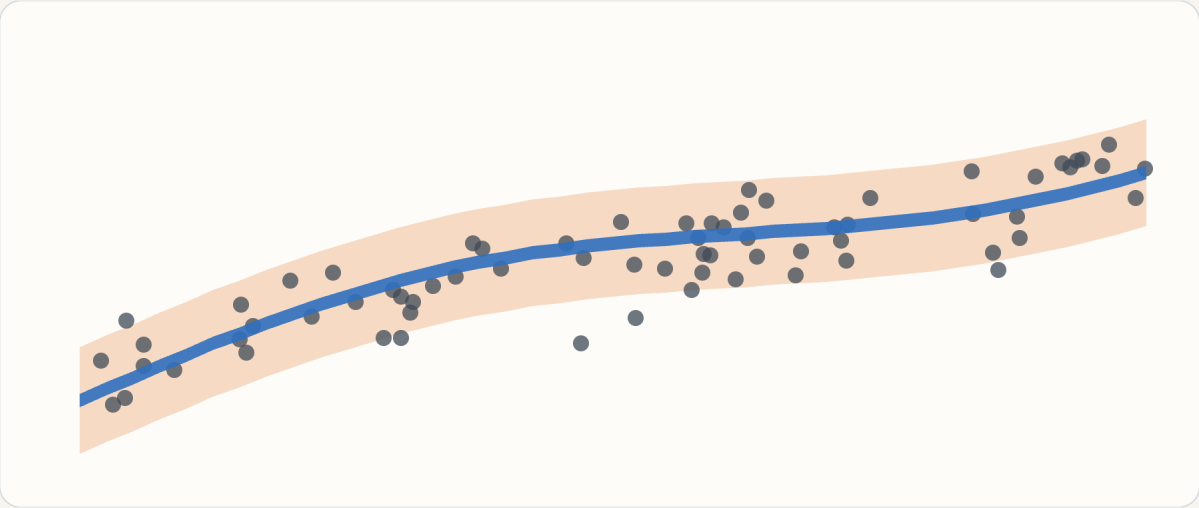
An anchor is tested if **either** end lies in the fold, trained on only if **neither** does. The warp must bridge a 2.7-min hole, so wiggles show up as error.

Pick λ from {0.1, 1, 10, …, 10^4} by the lowest median held-out error: **λ = 10**, error 0.111 min.

Your explanation of the leak is right. The TIC correlation is never used to choose λ; after selection it stays between 0.595 and 0.600 for every λ from 1 to 10^4.

Q13 · INTUITION

A thin thread for the line, a corridor for a peak



Sketch, not data. Dots: held-out anchors. Blue: the band for the map itself. Orange: the predictive interval for one compound.

Band: where is the line?

Refit 200 times on resampled samples. Thousands of anchors pin it: 0.044–0.054 min wide. Holds **15%** of held-out anchors.

Interval: where will a peak be?

Adds each peak's own scatter. Meant to hold 90%; holds **81%**. A chemist's search window in batch l.

You recalled the bands being insensitive to smoothness. The paper's finding is different: bands widen only weakly where direct anchors are sparse (Spearman ρ = −0.40).

Q13 · MATHS

Why a 90% interval catches only 81%

1 · Resample samples

In each batch, draw samples with replacement; sample i appears $m_{k,i}$ times. An anchor between two samples gets weight $m_{k,i} \cdot m_{l,i}'$ (0 if either is missing). Refit. Repeat 200 times.

2 · Band, then interval

The 5th–95th percentiles of the 200 maps give the band, half-width h. Add the pair's anchor scatter σ_{kl}:

$$\text{half-width} = \sqrt{(h^2 + (1.645 \cdot \sigma_{kl})^2)}$$

3 · The likely gap: bias

The smoothness penalty pulls every refit toward a smoother curve. Resampling shows how much the fits **vary**, not how far they are all **pulled**. The paper calls this "one likely reason"; it is not tested.

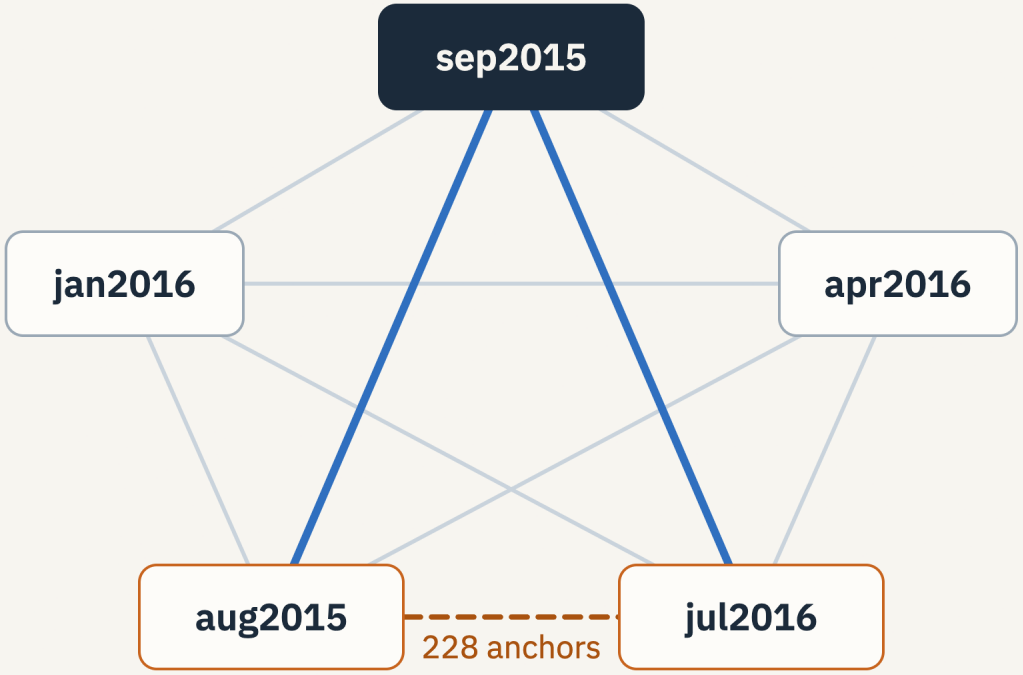
4 · The fix: calibrate

Widen the intervals until held-out anchors are covered at the stated rate, for example by conformal scaling. The paper recommends this before anyone relies on them.

1.645 is where a normal distribution leaves 5% in each tail, so ±1.645σ holds 90% if the scatter is Gaussian. Paper: uncertainty results; trust-align plans to test the calibration.

Q14 · INTUITION AND EVIDENCE

A weak link is held in place by the rest of the web



aug2015 has two samples, so the aug2015–jul2016 link is thin (orange). But both batches are tied to the clock through every other batch (blue: via sep2015). Neither can move without breaking those ties.

TIC correlation on that pair: pairwise 0.536, joint 0.598.

Delete a pair's own anchors and refit: changes ≤ 0.007 on 9 of 10 pairs; sep2015 × jan2016 drops 0.813 → 0.755.

Caveat: using only each batch's anchors against sep2015 (a star) stays within 0.016 of the full joint fit on pairs without sep2015. Much of the gain needs just one well-anchored reference.

CORRECTIONS · 1 OF 2

Where the chat answers were wrong: the fit

The chat said	What holds
$\|\|A\theta - y\|\|^2$ is strictly convex	Convex only: flat along “add one constant to every drift”. The gauge (and a tiny ridge) make each solve unique.
Huber’s loss is strictly convex	Convex, not strictly: past the threshold it is a straight line.
IRLS repeats until θ stops changing	A fixed schedule: 1 plain, 3 Huber and 4 Tukey solves.
With the gauge there is only one solution (your summary)	Each solve has one. The Tukey stage is not convex, so the final answer depends on where Huber leaves it.

Checked against the paper’s Method section and fit_joint in src/multibatch.py.

CORRECTIONS · 2 OF 2

Where the chat answers were wrong: the results

The chat said	What holds
A new gauge shifts the clock, which changes the penalty	A shift changes nothing. Gauges differ in the shape they give the clock, which the penalty does see.
The reference gauge was chosen from the benchmark	It was fixed by the selection rule; the comparison is a trade-off and the paper leaves it open.
Deleting a pair’s anchors gives virtually zero shift	≤ 0.007 on 9 of 10 pairs; sep2015 × jan2016 fell from 0.813 to 0.755.
Intervals under-cover because of smoothing bias	The paper’s “likely reason”, not a tested cause.

Checked against the paper’s results and discussion. The numbers the chat quoted (0.116, 0.277, 0.0023, 81%, 15%, $\lambda = 10$) all match the paper.

THE WHOLE METHOD

Six ideas, each answering one of your questions

1 · One clock

One warp per batch; maps go via the clock, so every path agrees.

2 · Linear misses

Warp forward, hat basis, increments: each anchor is one row of A.

3 · A box

Slope in [0.5, 2] becomes bounds on each δ: bounded least squares.

4 · A gauge

Pins the clock so each solve has one answer; its choice is open.

5 · Robust, then tuned

Huber then Tukey weights; λ = 10 from time-blocked folds.

6 · Honest uncertainty

Bootstrap bands and intervals: 81% for 90%, so calibrate first.

Borrowing strength comes free from idea 1: a thin pair is held by every path through the other batches.